

MACAPA, Brazil

WMO: 820980

Lat: 0.0451S

Lon: 51.1099W

Elev: 42

StdP: 14.67

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	73.0	73.3	69.8	109.8	88.7	71.2	115.3	85.5	19.2	84.6	16.8	85.1	4.2	0	0.390

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10	17.0	95.4	79.6	95.2	79.6	94.7	79.7	82.5	89.9	81.8	89.5	81.2	89.2	11.5	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.9	160.7	83.7	80.5	158.4	83.7	79.2	151.7	83.5	46.2	89.0	45.6	90.8	45.0	90.5	86.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.1	16.1	14.4	DB	71.1	96.1	4.6	1.6	67.8	97.2	65.1	98.2	62.5	99.1	59.1	100.3	(4)
(5)			WB	70.1	83.8	4.4	1.6	66.9	85.0	64.4	85.9	61.9	86.7	58.8	87.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	83.3	82.1	81.3	81.5	82.0	82.8	83.0	83.2	84.4	84.9	85.1	85.0	83.9
	DBStd	2.47	2.56	2.62	2.56	2.36	2.27	1.96	1.87	1.66	1.43	1.54	1.65	2.08
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	12153	996	877	975	960	1017	991	1030	1067	1047	1090	1051	1052
	CDD65	6678	531	457	510	510	552	541	565	602	597	625	601	587
	CDH74	70664	4994	3897	4383	4570	5299	5445	5843	7013	7358	7843	7503	6516
	CDH80	29442	1828	1248	1354	1457	1889	2122	2430	3219	3546	3856	3648	2845
Wind	WSAvg	7.2	7.1	6.5	6.2	5.7	5.5	5.6	5.9	7.5	9.0	9.4	9.3	8.3
Precipitation	PrecAvg	91.0	9.9	13.5	15.0	14.0	11.0	9.6	7.0	3.5	1.0	0.9	1.9	5.4
	PrecMax	105.3	18.7	26.0	24.6	22.6	16.9	15.7	12.1	7.6	3.0	3.5	8.4	15.1
	PrecMin	68.9	2.3	6.9	5.0	7.5	5.3	4.8	3.9	0.6	0.0	0.0	0.0	0.2
	PrecStd	9.1	4.0	4.4	4.0	3.9	3.0	2.7	2.3	1.8	0.9	1.0	2.4	3.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	94.7	93.4	93.5	94.7	95.2	95.2	95.3	96.7	96.9	97.0	96.9	96.6
		MCWB	79.5	80.5	81.0	80.5	80.7	80.2	79.8	79.6	79.5	79.3	78.8	79.2
	2%	DB	92.8	91.5	91.5	91.7	93.2	93.3	93.5	95.0	95.2	95.3	95.3	94.9
		MCWB	79.8	80.0	80.2	80.4	81.0	80.3	80.0	79.8	79.7	79.4	79.3	79.4
	5%	DB	91.0	89.6	89.4	89.7	91.2	91.5	91.8	93.4	93.6	94.7	94.7	93.2
		MCWB	79.5	79.4	79.5	79.7	80.4	79.9	79.7	80.0	79.6	79.4	79.4	79.8
	10%	DB	88.2	87.5	87.4	87.6	89.2	89.7	90.0	91.7	92.9	93.1	93.1	91.3
		MCWB	78.9	79.1	79.1	79.5	79.7	79.3	79.1	79.5	79.5	79.2	79.3	79.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	81.8	82.5	82.5	82.6	83.0	82.6	82.5	82.7	82.2	82.0	82.0	82.4
		MCDB	89.7	89.4	89.6	89.3	90.2	89.2	89.9	90.9	91.4	91.9	90.5	88.7
	2%	WB	80.8	81.0	81.0	81.3	81.8	81.5	81.4	81.7	81.1	80.9	81.0	81.1
		MCDB	88.7	87.9	87.4	87.6	88.7	88.7	89.3	90.5	91.0	91.3	90.6	89.3
	5%	WB	80.0	80.1	80.3	80.6	80.9	80.7	80.6	80.8	80.6	80.3	80.4	80.5
		MCDB	87.4	86.4	86.1	86.7	87.5	88.3	88.9	90.1	90.8	90.7	90.5	89.3
	10%	WB	79.3	79.2	79.5	79.8	80.2	80.0	79.9	80.1	79.9	79.6	79.7	79.7
		MCDB	86.0	84.7	84.6	85.1	86.2	86.9	87.5	88.9	89.6	89.8	89.6	88.1
Mean Daily Temperature Range	5% DB	MDBR	13.9	12.8	12.6	12.8	13.8	14.9	15.9	16.7	17.2	17.0	16.7	15.6
		MCDBR	16.4	15.4	15.6	15.6	16.6	17.1	17.5	18.2	18.3	18.4	18.2	17.8
		MCWBR	5.3	5.5	5.5	5.3	5.6	5.8	6.1	5.8	5.5	5.3	5.2	5.4
	5% WB	MCDBR	14.8	14.2	14.1	14.0	15.0	15.7	16.6	17.0	17.5	17.4	17.1	16.3
		MCWBR	5.5	5.7	5.5	5.5	5.7	6.0	6.4	6.3	5.8	5.7	5.7	5.6
Clear-Sky Solar Irradiance	taub	0.409	0.434	0.425	0.403	0.414	0.412	0.416	0.445	0.485	0.478	0.468	0.448	
	taud	2.508	2.437	2.471	2.558	2.501	2.505	2.484	2.381	2.230	2.250	2.281	2.371	
	Ebn at Noon	292	285	286	286	275	272	272	270	265	270	272	279	
	Edh at Noon	35	38	37	33	34	33	34	39	46	46	44	40	
All-Sky Solar Radiation	RadAvg	1377	1270	1290	1323	1423	1566	1677	1864	1916	1819	1699	1480	
	RadStd	95	79	77	64	87	102	58	68	62	81	84	111	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+1.48	+0.76	N/A	+2.30	+1.04	+1.46	N/A	N/A	+785
(47) Regional (0 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+781

Nomenclature: See separate page